



A
PROJECT REPORT
ON

'Online Examination System'

Submitted to
SAVITRIBAI PULE PUNE UNIVERSITY

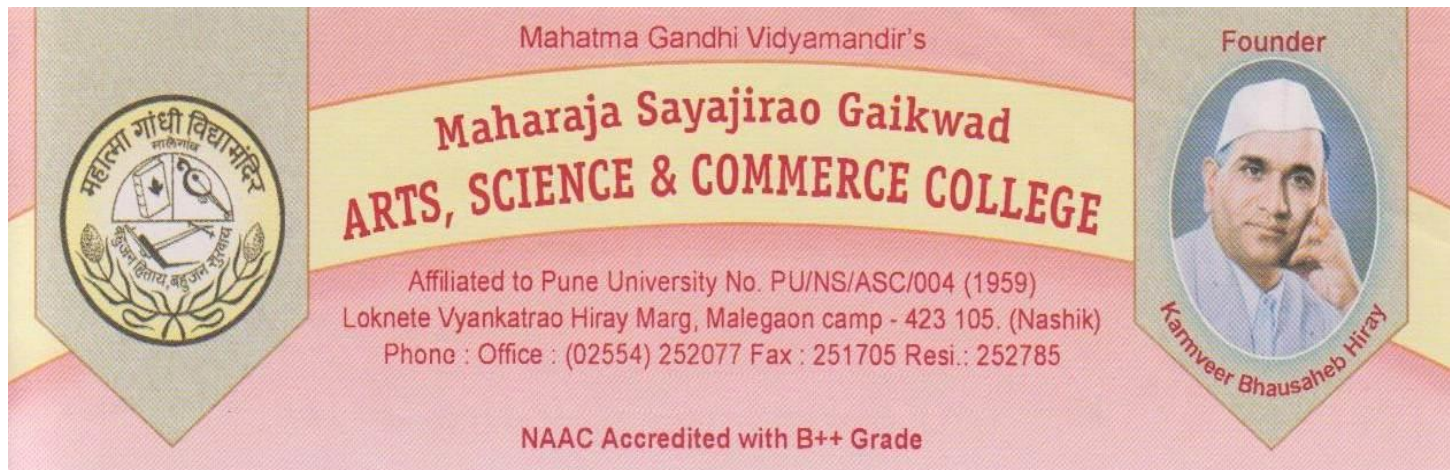
MGV's
M. S. G. Arts, Science and Commerce College,
Malegaon Camp

'DEPARTMENT OF COMPUTER SCIENCE'

In the fulfilment for the degree
Of
Bachelor of Science (Computer Science)

By
Thakare Pranoti Kamalalar (19)
Chavan Mamta Eknath (20)
Ahire Nalini Ashok (21)
Musale Vaishnavi Kishor (54)

2022-2023



CERTIFICATE

This is to certify that Mr. / Miss **Thakare Pranoti Kamalakar (19), Chavan Mamta Eknath (20), Ahire Nalini Ashok (21), Musale Vaishnavi Kishor (54)** have completed their project on “**ONLINE EXAMINATION SYSTEM**” satisfactorily in the partial fulfilment of class **B. Sc. (Comp. Sci.)**, an Under Graduate Degree course affiliated to Savitribai Phule Pune University for the academic year 2022-2023.

Project Guide

D. J. Deore

Internal Examiner

External Examiner

ACKNOWLEDGEMENT

We take this opportunity to thank all those openhanded persons who rendered their full support to our work.

We express our gratitude to Principal Dr. S. N. Nikam for the cooperation extended to the successful completion of the project. In addition, with a lot of happiness, we are expressing gratitude to Mr. D. J. Deore, Head of the Computer Science department, for his timely and kind help. Also, his inspiration up to the last moment had made things possible in a planned manner. We also thank our teachers for their guidance and for providing us with the most essential materials required for the completion of this project. We are very thankful to our guide for his timely guidance.

Finally, we thank each and every one who helped to complete our project work with their cordial support.

Project Team Members

Thakare Pranoti Kamalakar	(Roll No. 19)
Chavan Mamta Eknath	(Roll No. 20)
Ahire Nalini Ashok	(Roll No. 21)
Musale Vaishnavi Kishor	(Roll No. 54)

ABSTRACT

Project Title: Online Examination System

Abstract:

In online examination system, exams are conducted through specific institute. This process is done by using internet. It needs less hard work for conducting exams. This system checks the activities and performance of student. Compare to the earlier situation, it was necessary to prepare the question papers physically and distribute them to the examination centres. But with this system, question papers can be created in digital format and distributed as per requirement.

Due to this system, the chances of malpractices as well as question paper leakage can be reduced to great extent. This system makes it very easy to produce the result and can save a lot of work related to the answer sheet. It also helps environment by saving paper. One more benefit of this system is that the student can easily participate in exams from anywhere. Most importantly, the online examination system allows the student to take the exam at her/his convenience.

Project Development Specification:

- **Frontend:** PHP, CSS, HTML, JavaScript
- **Backend:** PostgreSQL

Project Members:

1. Thakare Pranoti Kamalakar (19)
2. Chavan Mamta Eknath (20)
3. Ahire Nalini Ashok (21)
4. Musale Vaishnavi Kishor (54)

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1. Introduction to Proposed System

a. Motivation:

An online examination system is a system that establishes a network between students and institutions. This system caters to the needs of institutions conducting online exams. Institutions must apply separately for online examination. This system will work to handle multiple services so that records of students are safe and properly taken care of. This system will save time and paper work and conduct the exams smoothly. In this system students can see the result only after the examination.

In today's world many institutions use this online examination system. This system supports short type of questions. This system allows the institutes to put the questions they want in the exams. Also, these questions are displayed as exam for eligible students. This examination system fulfils the requirements of the coaching centres as well as institutes to conduct online exams. Institutes just must register an application and the list of students that appear in exam. Online Exam System is more convenient than traditional examination system.

b. Problem Statement:

The problem with the current system is that student take exams manually. This outdated system will take long time utilization, the manual procedure used for conducting exam is much time-consuming process. Since covid-19 situation the existing system have many drawbacks like its time consuming. Results are not accurate since calculation is done manually.

Online Examination system saves the exams information in a database, and this make it an easier way to give exams. Students can give exam in a totally automated system.

c. Objectives:

1. Evaluates the student's ability of learning.
2. When the student starts exam, time will automatically and show the student how much time is left.
3. User friendly interface.
4. Time flexibility.
5. Automatically checked answers.
6. That system keeps our data safe and secure.

d. Project Scope:

1. Online Examination System are used anywhere any time as it is a web-based system.
2. The candidates can enrol through the client computers with their seat numbers that given to them.
3. The main purpose of this system is candidate can use fully automated system that save the time as well as it save

e. Limitations:

1. It is not applicable for unregistered students.
2. This system does not access without internet.
3. It must require laptop or android device.
4. Technical problems or connection error may occur.

2. System Analysis

a. Existing System:

The Existing System is the entry of humans into the digital world. The information of students already registered in this system is kept safe. One month registration details of students are required to be maintained but it is difficult to do the details manually. This requires a lot of manpower. It also causes loss of money and time. The Existing examination system is the solution to all these problems. The present exam is also known as the virtual exam. It also enables the students who are far away from the exam centre to conduct the exam using Internet. But students are expected to answer questions on this exam system. As in the conventional examination system, the existing system is given a fixed time period. So, the existing online exam gets auto-submitted after exam time is over. After submission of exam, we cannot login it again.

b. Scope & Limitations:

1. Offline exams are very time consuming.
2. In this existing system the exams are done only manually.
3. In the existing examination system, we write exam through paper and it is wastage of time to print question papers in the form of hardcopy.
4. It is very difficult to save or maintain question paper from damage.
5. In existing system it is necessary to reach on time at the exam centre.
6. Printing question paper in this system takes a lot of time and manpower as well as it takes lot of time to show results.
7. Due to change in environment, it becomes difficult for the students who are far away to reach the exam centre.
8. In existing system, there is lot of wastage of time and money in transportation for students those who are come from another city.

Stakeholders:

Team Members :-

1. Thakare Pranoti Kamalakar (Roll No. 19)
2. Chavan Mamta Eknath (Roll No. 20)
3. Ahire Nalini Ashok (Roll No. 21)
4. Musale Vaishnavi Kishor (Roll No. 54)
5. College.
6. Seniour college Student.

3. Requirement Analysis

a. Functional Requirements:

1. The teacher will be able to add new questions in the exam.
2. The examination system will maintain different subject for exam.
3. The teacher can decide start time and end time of exam.
4. The system will be able to analyse exam.
5. The teacher will be able to edit or delete a question.
6. The supervisor will be able to select maximum number of questions to be include in exam.
7. User will be required to enter login id and password.

b. Performance Requirements:

1. The database should be able to accommodate a minimum of 10,000 records of students.
2. The software shall support use of multiple users at a time.
3. There are no other specific performance requirements that will affect development.
4. The system shall take as less time as possible to provide service to the administrator or the staff.

c. Social Requirements:

1. Some of the factors that are identified to protect the software from accidental or malicious access, use, modification, destruction or disclosure . Keep specific log or history data sets.
2. Assign certain functions to different modules.
3. Restrict communications between some areas of the program.
4. Check data integrity for critical variables.
5. Later version of the software will incorporate encryption techniques in the user authentication process.

d. Data Collection:

1. User Id and password are important for students to give online exam.
2. Similarly, the institute which has conducted the exam also needs user id and password.
3. A question bank is provided on the examination portal to facilitate the study of the students.

e. Software / Hardware Specifications:

Hardware Requirements:

For the server:

1. Processor: Standard processor with a speed of 1.6 GHz or more.
2. RAM: 256 MB RAM or more
3. Hard Disk: 3 GB or more.

For the Client:

1. Processor: Standard Processor with a speed of 1.6 GHz or more.
2. RAM: 256 MB RAM or more.
3. Hard Disk: 3 GB or more.

Software Requirements:

For the Server:

1. Operating System: Linux
2. Frontend: PHP, CSS, JavaScript
3. Backend: PostgreSQL (Version 8)
4. Server: Apache 2, Wamp Server, Zamp Server (Any)

For the Client:

1. Operating System: Linux
2. Browsers: Any

4. System Design

a. Data Dictionary:

Sr No.	Table Name	Description
1.	User Table	Student table contains first name, middle name, last name, user name, email, password, age, mobile no., address
2.	Test Table	Test table contains test id, subject id, test name, total no of questions.
3.	Subject Table	Subject Table contains subject id and subject name.
4.	Question Table	Question Table contains question id, test id, question description, option1, option2, option3, option4, true option.
5.	User Answer Table	User answer table contains session id, test id, question description, option1, option2, option3, option4, true answer, and your answer.
6.	Result Table	Result Table contains login, test id, test date, and score

i. User Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	u_id	Serial	-	Primary Key	User ID
2.	f_name	character varying	30	Not Null	First Name
3.	m_name	character varying	30	Not Null	Middle Name
4.	l_name	character varying	30	Not Null	Last Name
5.	u_name	character varying	30	Not Null	User id
6.	u_email	character varying	30	Not Null	Email
7.	u_pass	character varying	30	Not Null	password
8.	u_age	Integer	-	Not Null	Age
9.	u_mob	Bigint	-	Not Null	Mobile No.
10.	u_adr	character varying	30	Not Null	Address

ii. Test Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	test_id	Serial	-	Primary Key	Test ID
2.	sub_id	Integer	-	Not Null	Subject id
3.	test_name	character varying	30	Not Null	Test name
4.	tot_que	Integer	-	Not Null	Total no. of questions

iii. Subject Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	sub_id	Serial	-	Primary Key	Subject ID
2.	sub_name	character varying	30	Not Null	Subject Name

iv. Question Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	que_id	Serial	-	Primary Key	Question ID
2.	test_id	Integer	-	Not Null	Test id
3.	que_desc	character varying	80	Not Null	Question description
4.	ans1	character varying	50	Not Null	Option1
5.	ans2	character varying	50	Not Null	Option 2
5.	ans3	character varying	50	Not Null	Option 3
6.	ans4	character varying	50	Not Null	Option 4
7.	true_ans	Integer	-	Not Null	True answer

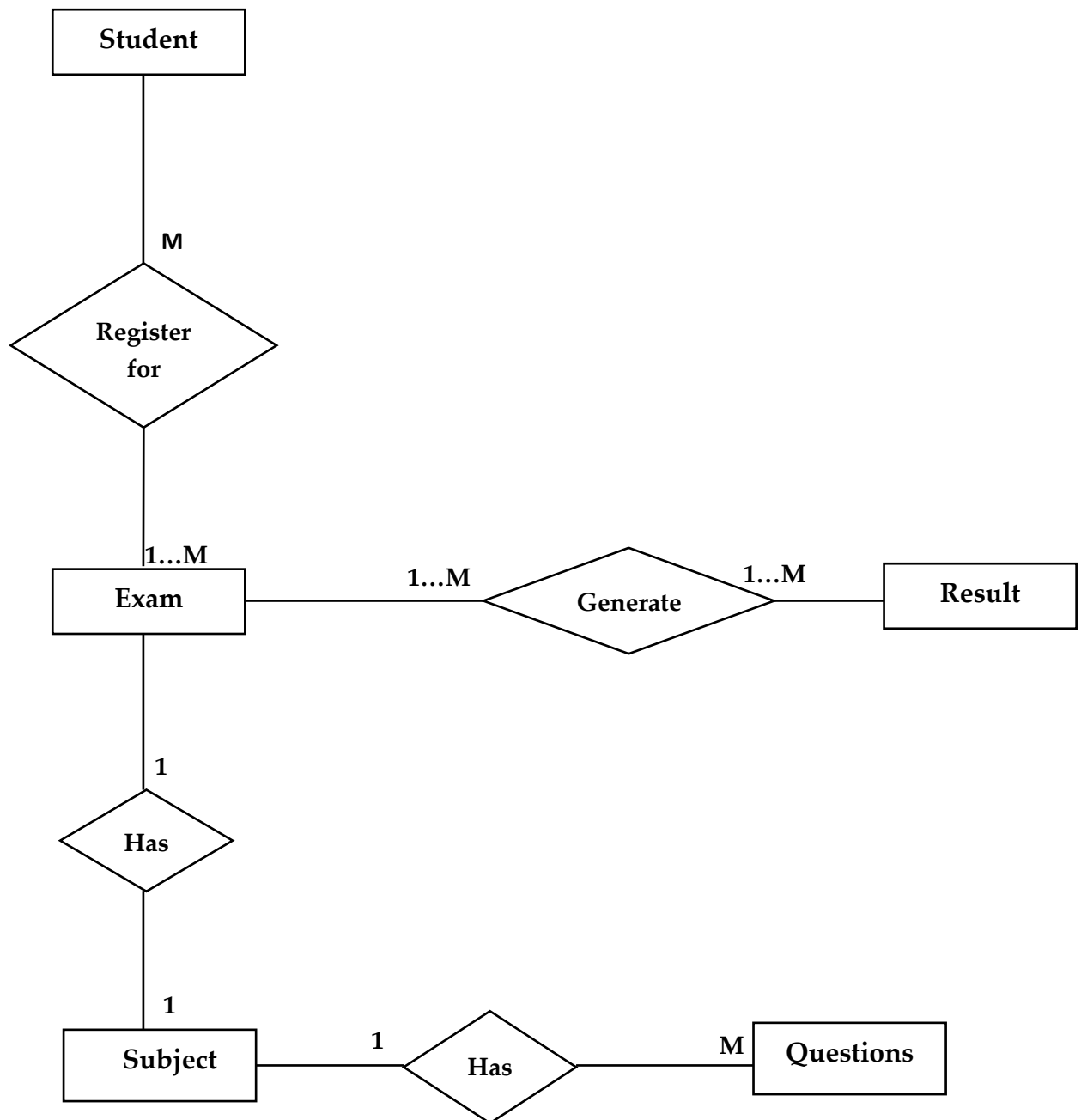
v. User Answer Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	sess_id	Serial	-	Primary Key	Session id
2.	test_id	Integer	-	Not Null	Test ID
4.	que_des	character varying	80	Not Null	Question description
5.	ans1	character varying	50	Not Null	Option1
6.	ans2	character varying	50	Not Null	Option 2
7.	ans3	character varying	50	Not Null	Option 3
8.	ans4	character varying	50	Not Null	Option 4
9.	true_ans	Integer	-	Not Null	True answer
10.	your_ans	Integer	-	Not Null	Your answer

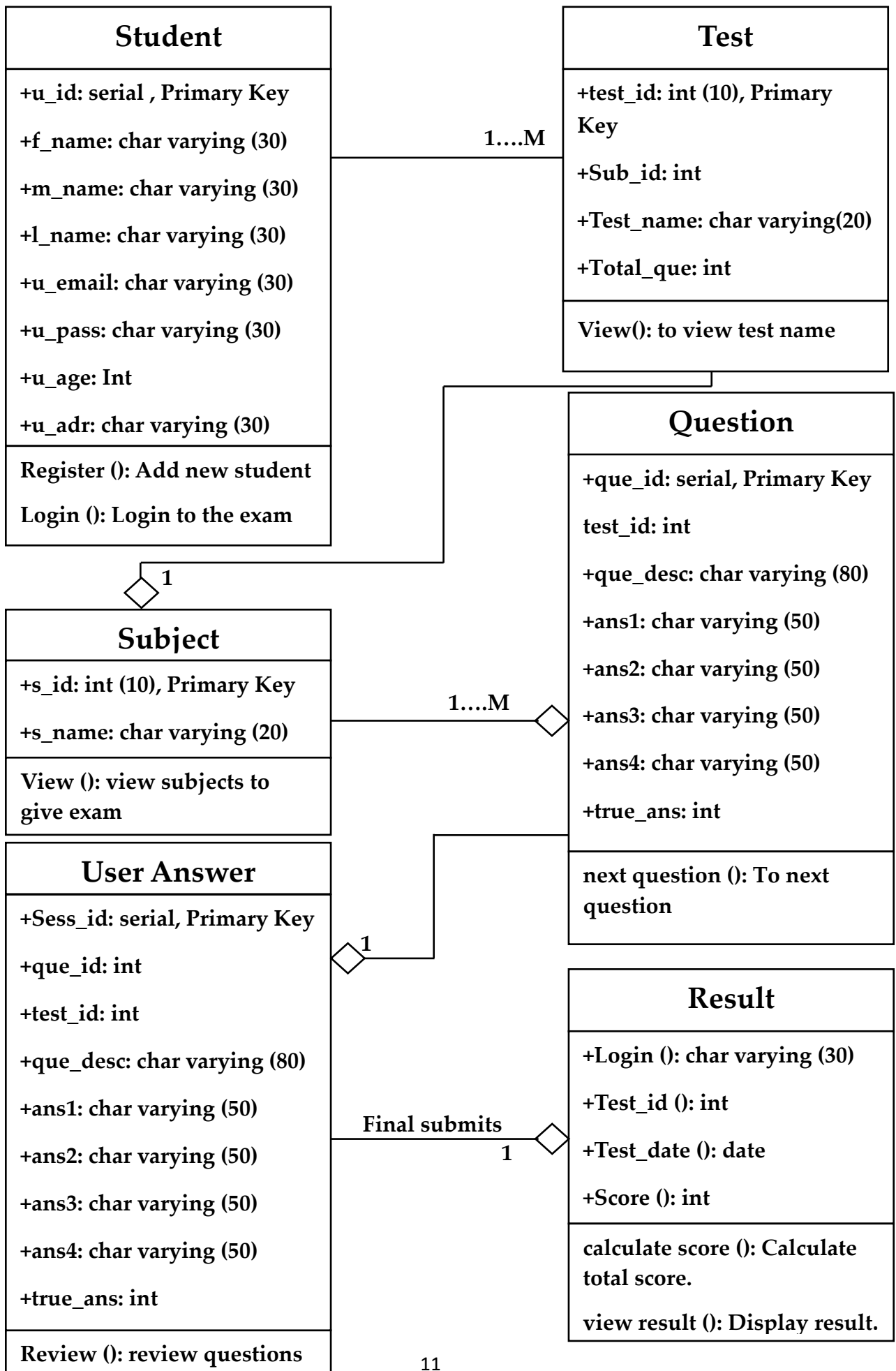
vi. Result Table:

Sr No.	Attributes	Data Type	Size	Constraints	Description
1.	login	Character Varying	30	Not Null	Login
2.	test_id	Integer	-	Not Null	Test id
3.	test_date	Date	-	Not Null	Test Date
4.	score	Integer	-	Not Null	Result Score

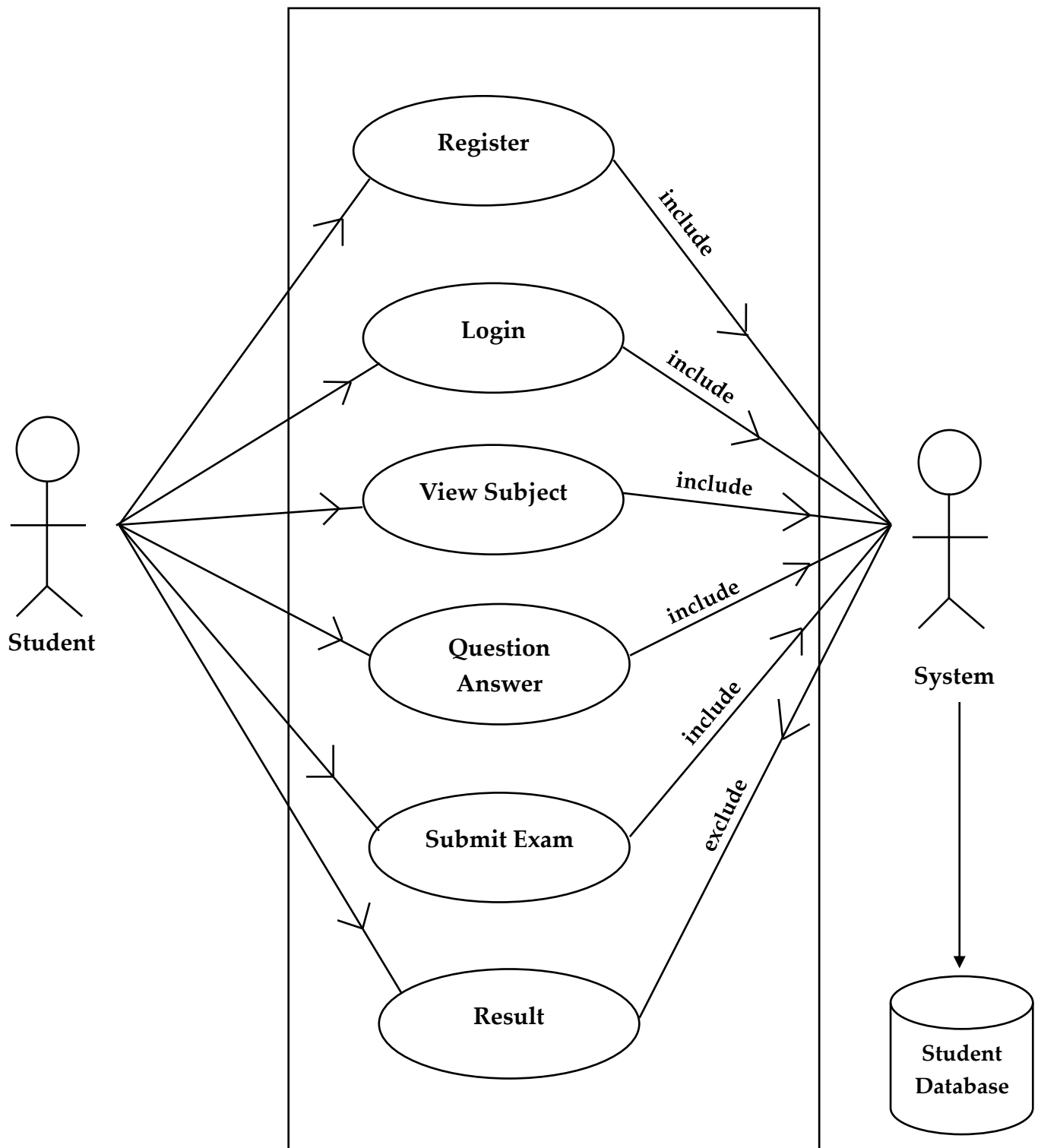
1) Entity Relationship Diagram:



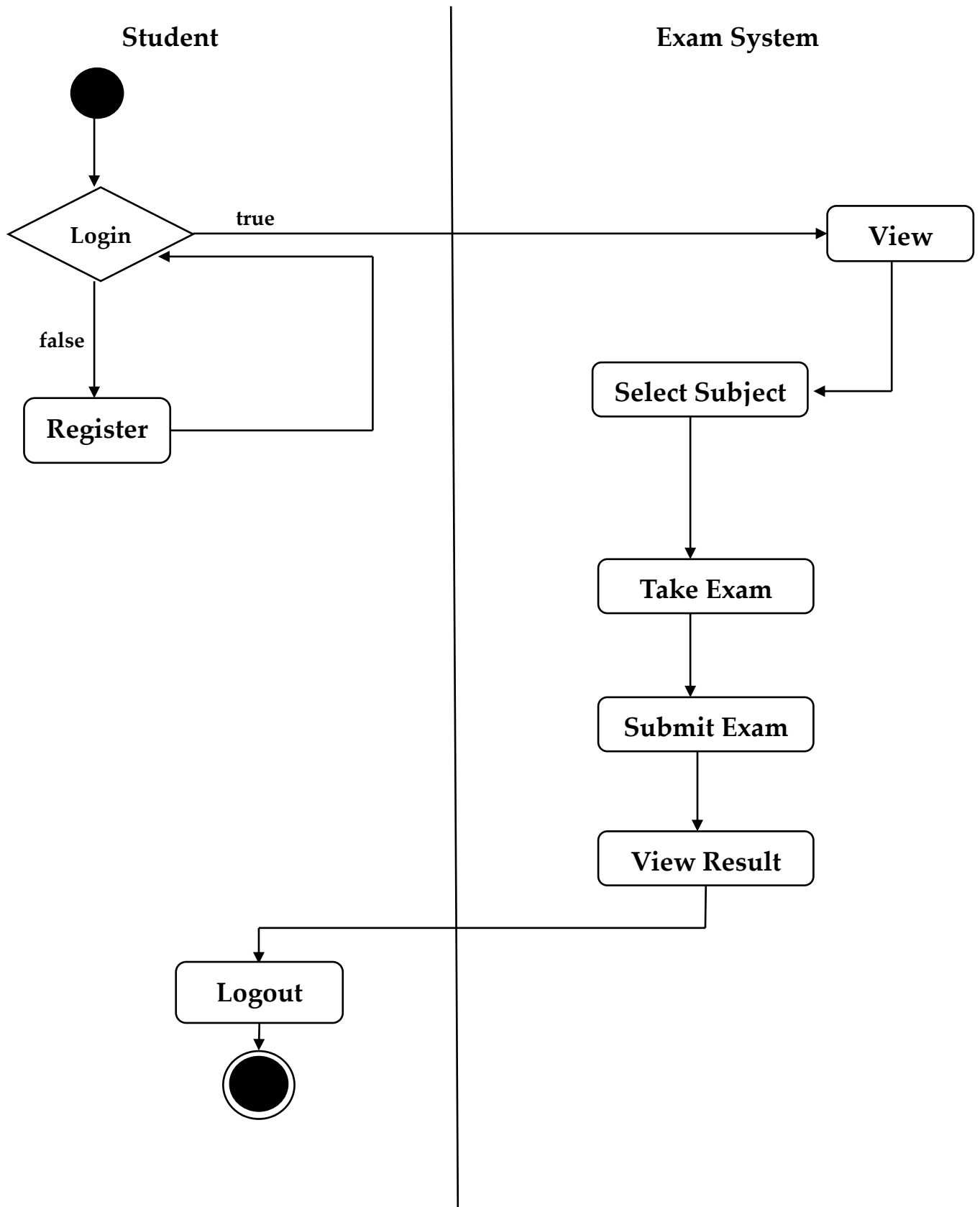
2) Class Diagram:



3) Use Case Diagram:

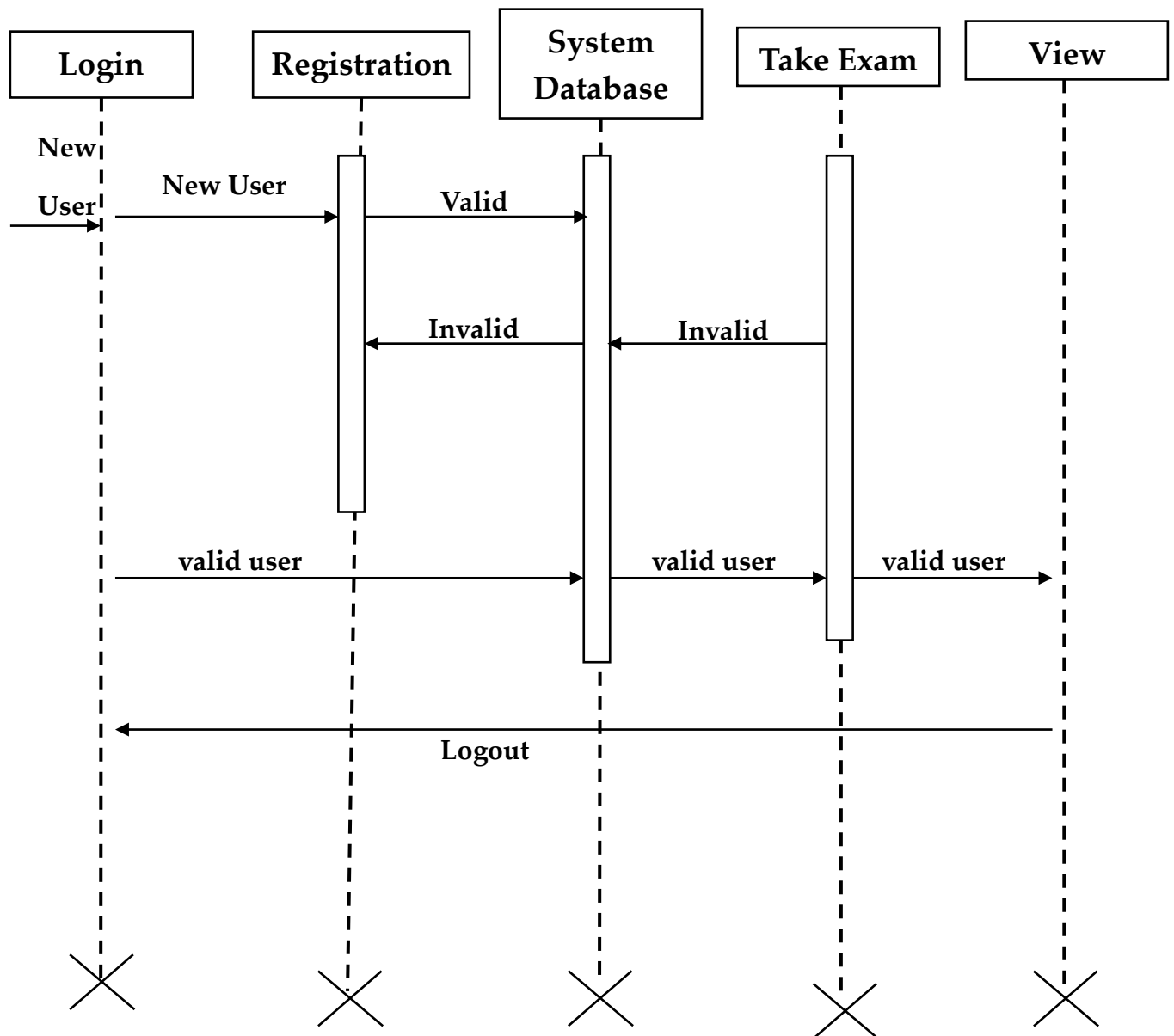


4) Activity Diagram:

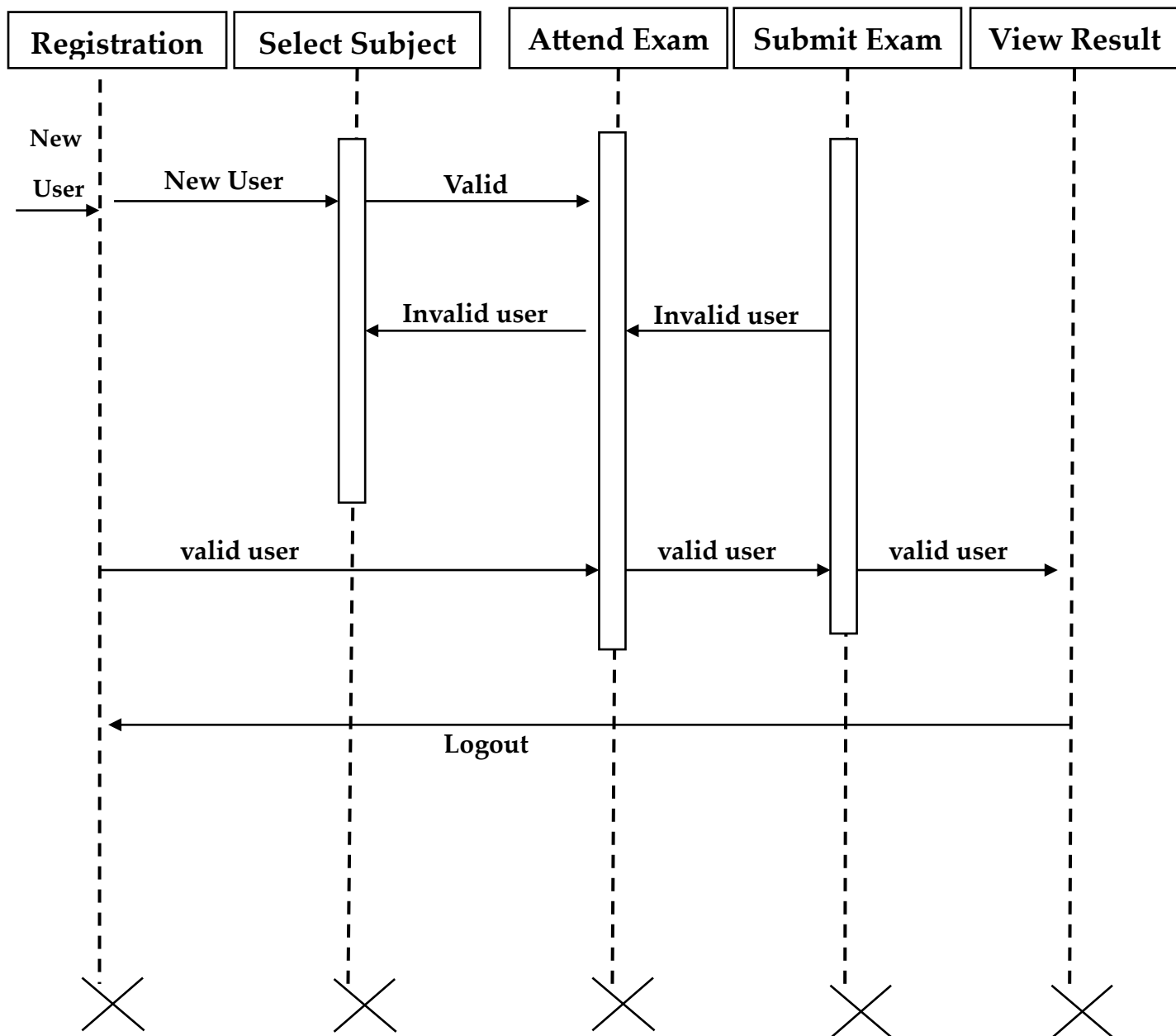


5) Sequence Diagram:

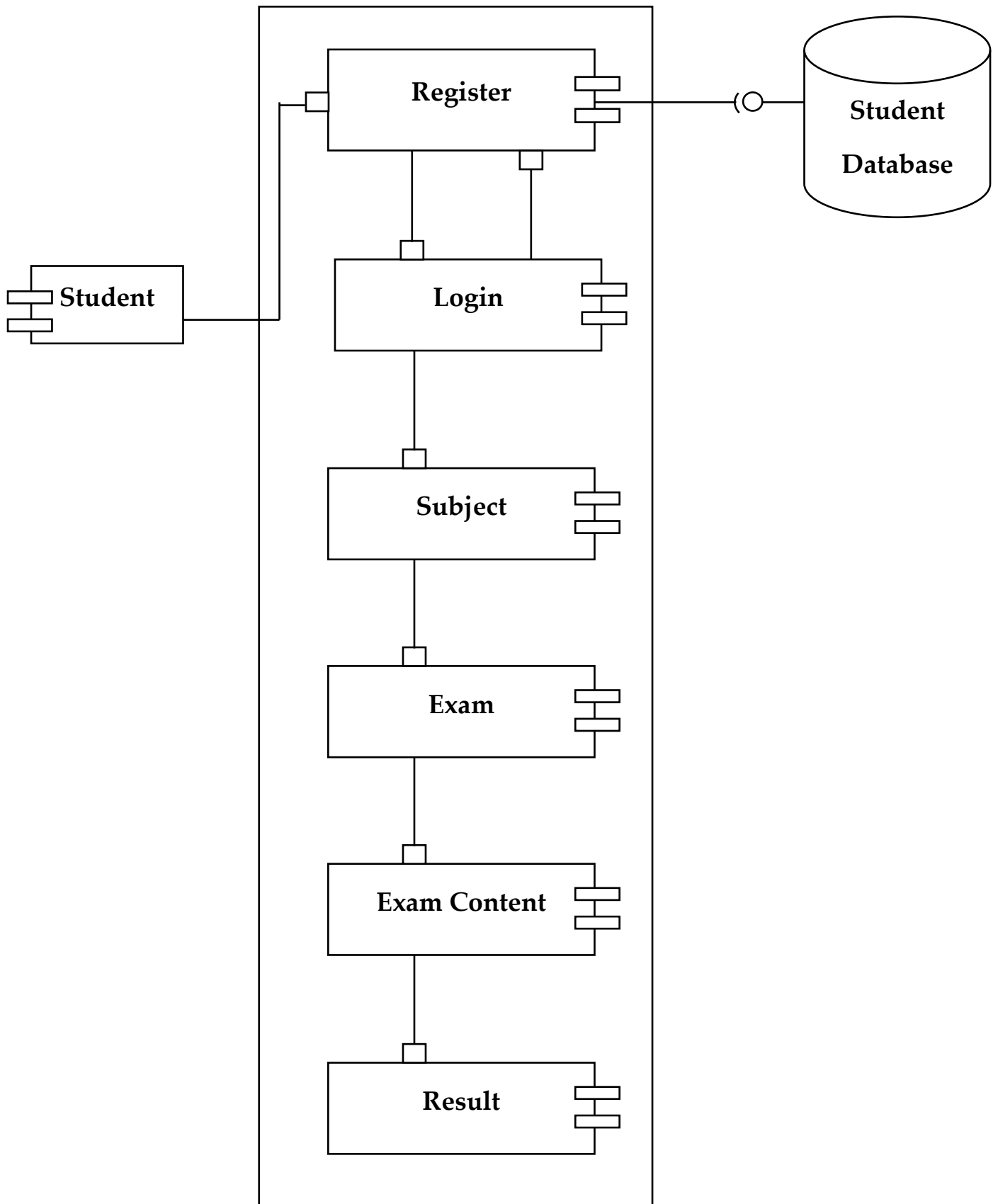
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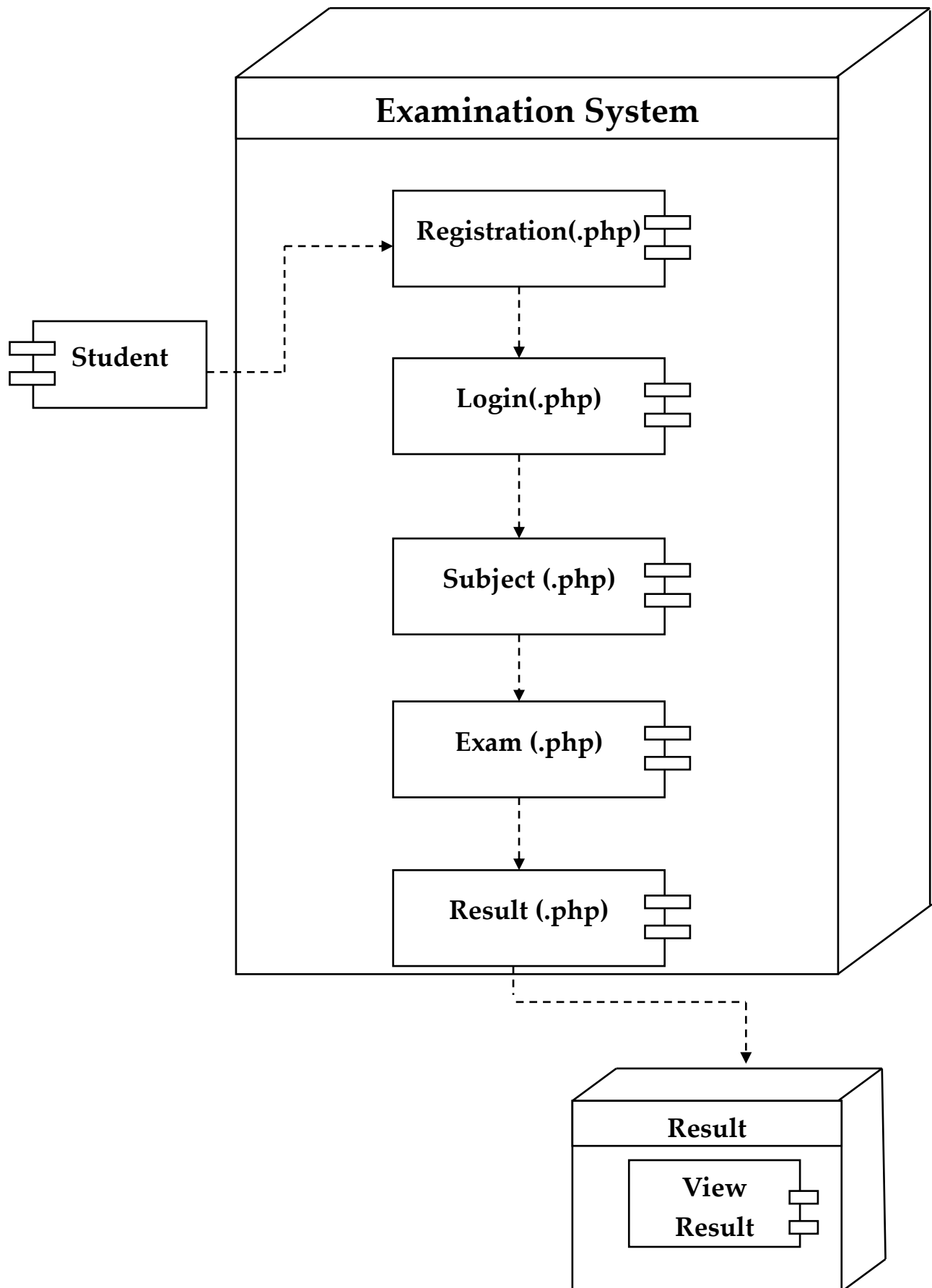
For Exam:



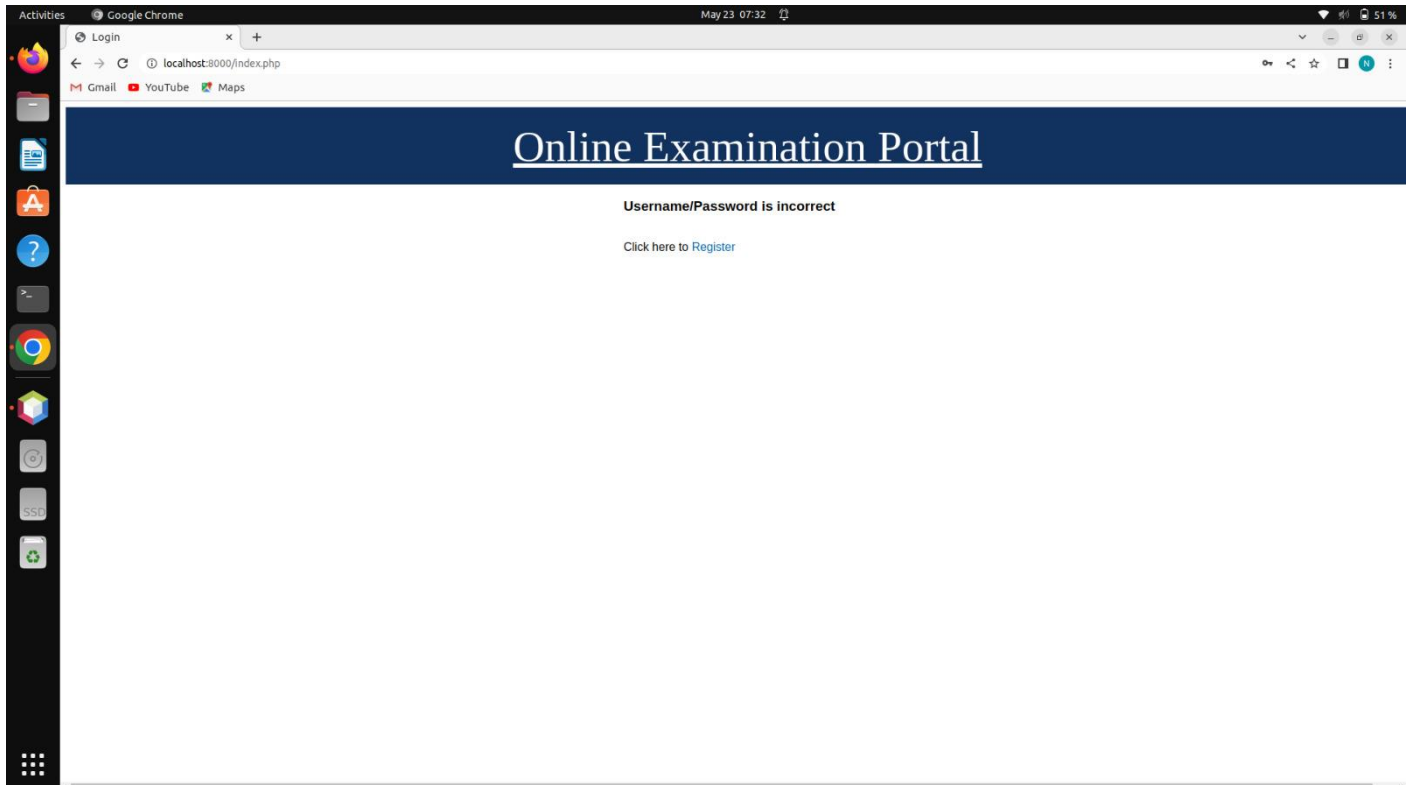
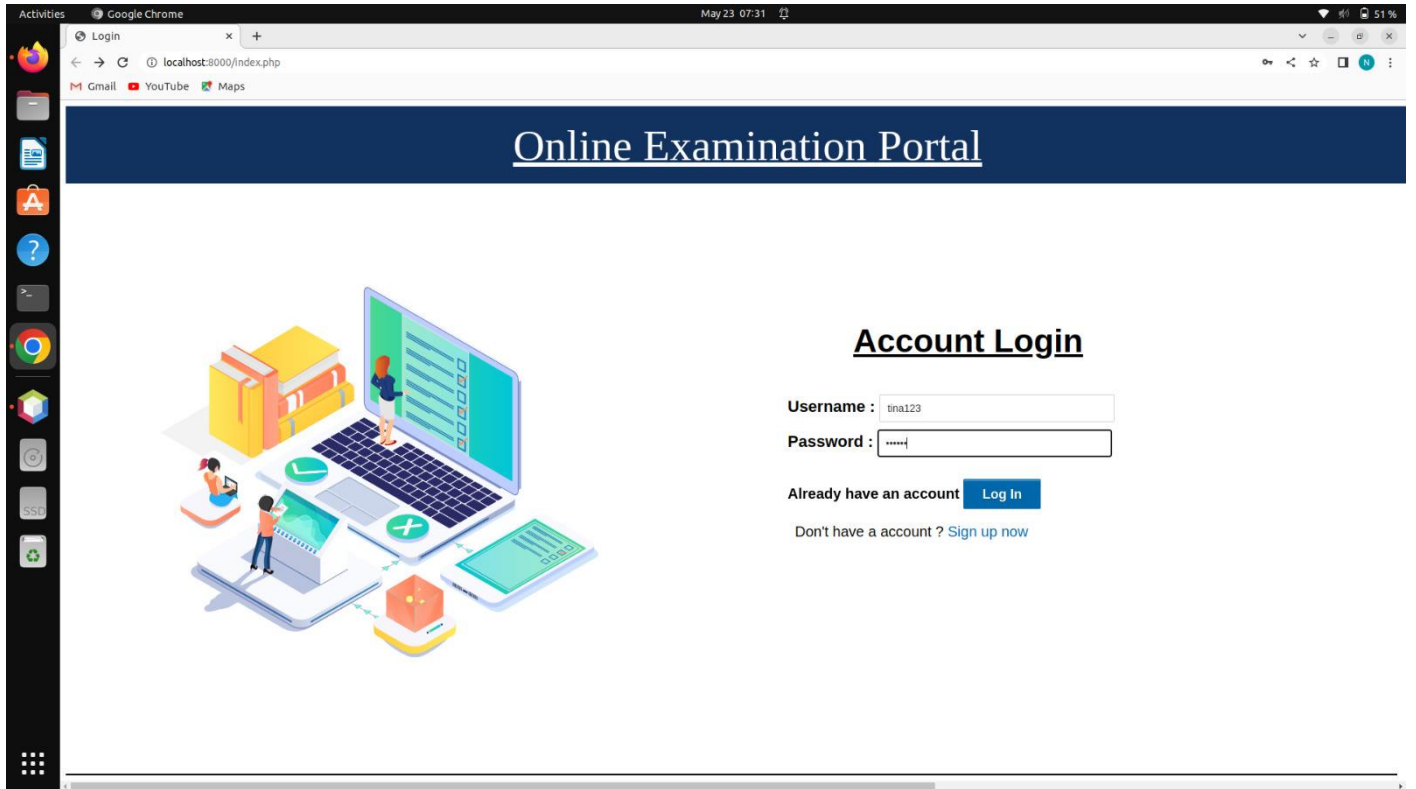
6) Component Diagram:



7) Deployment Diagram:



5.Implementation System



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Registration x +

localhost:8000/Registration.php

Gmail YouTube Maps

Online Examination Portal



Registration

Please fill in your credentials to Register

First Name :

Middle Name :

Last Name :

User Name :

Email-id :

Password :

Age :

Mobile No. :

Activities Google Chrome May 23 07:33 51%

Registration x +

localhost:8000/Registration.php

Gmail YouTube Maps



Registration

Please fill in your credentials to Register

First Name :

Middle Name :

Last Name :

User Name :

Email-id :

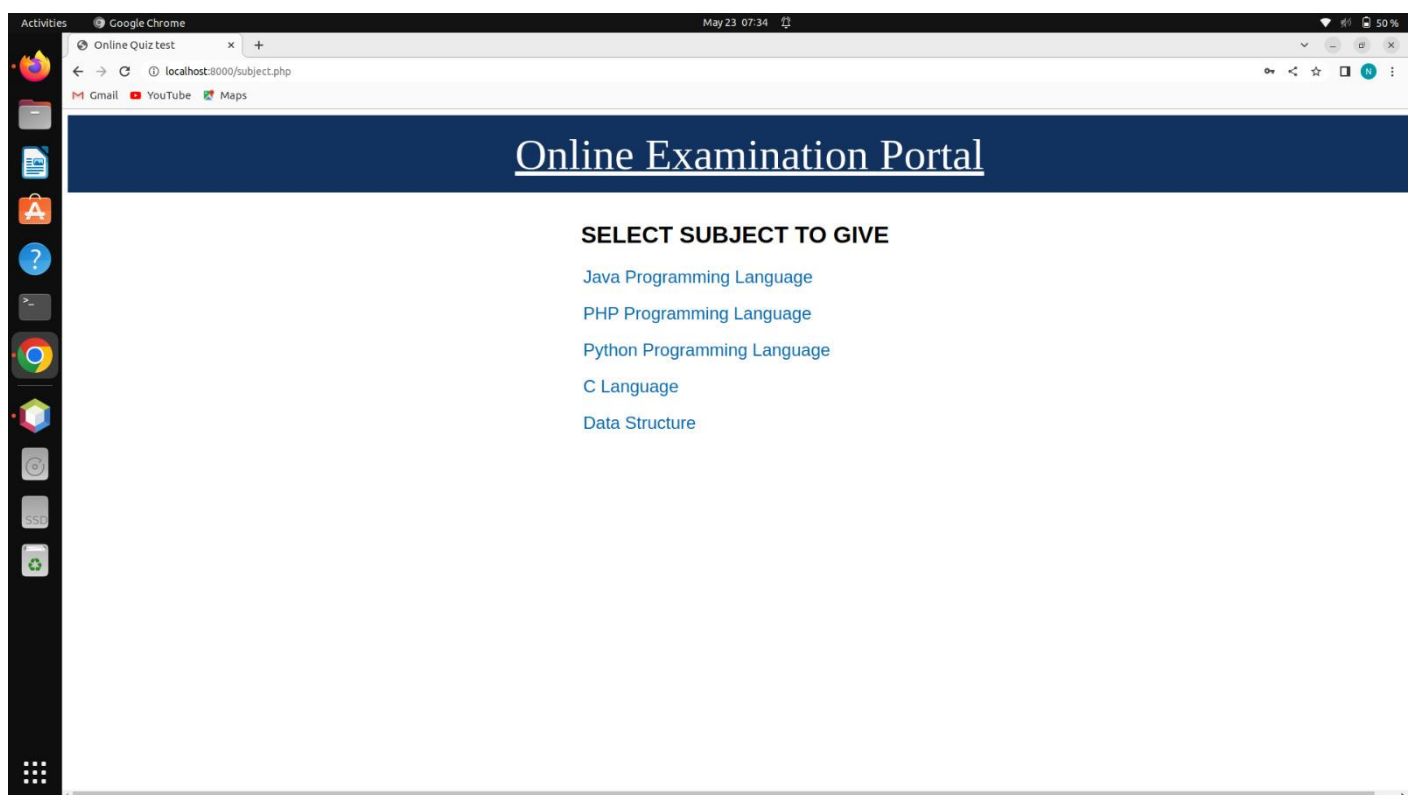
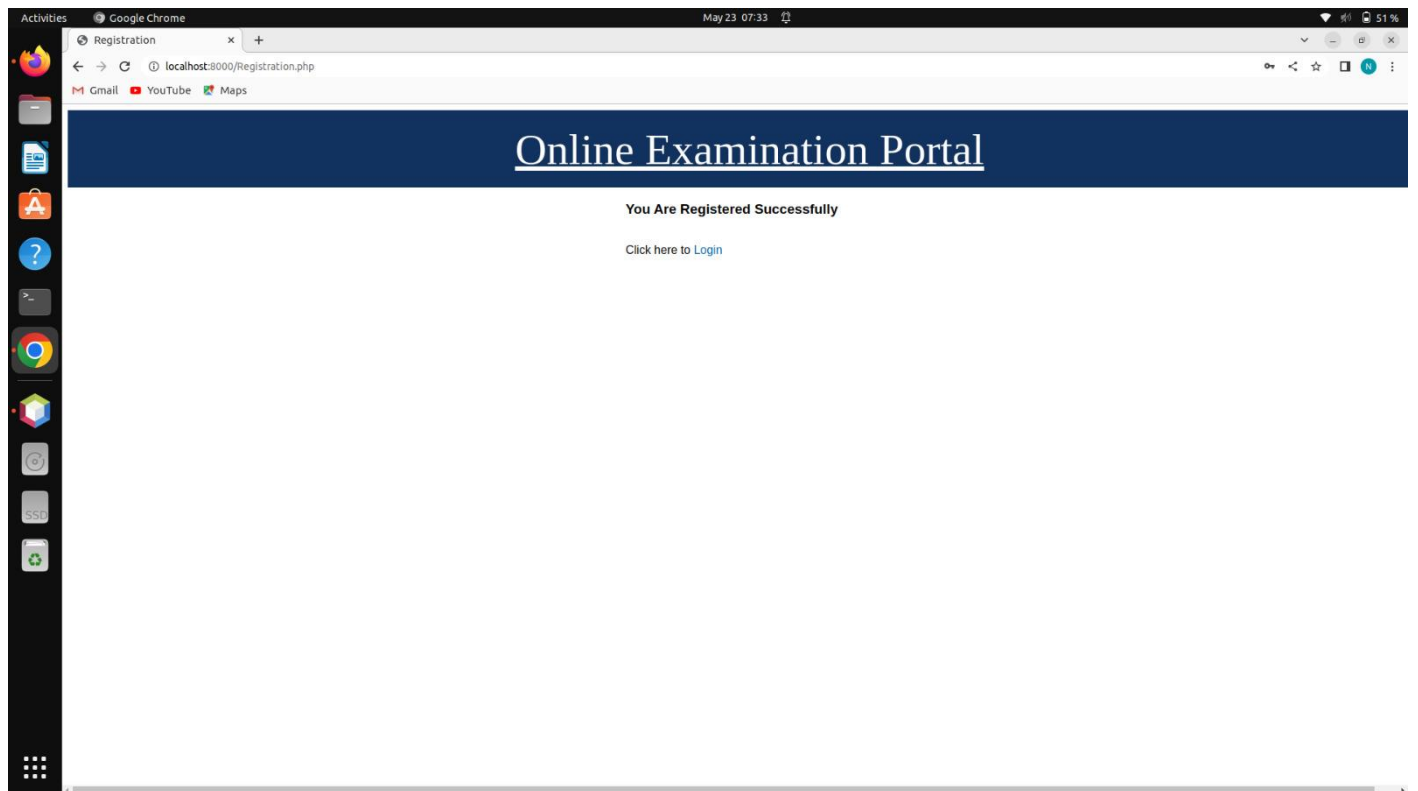
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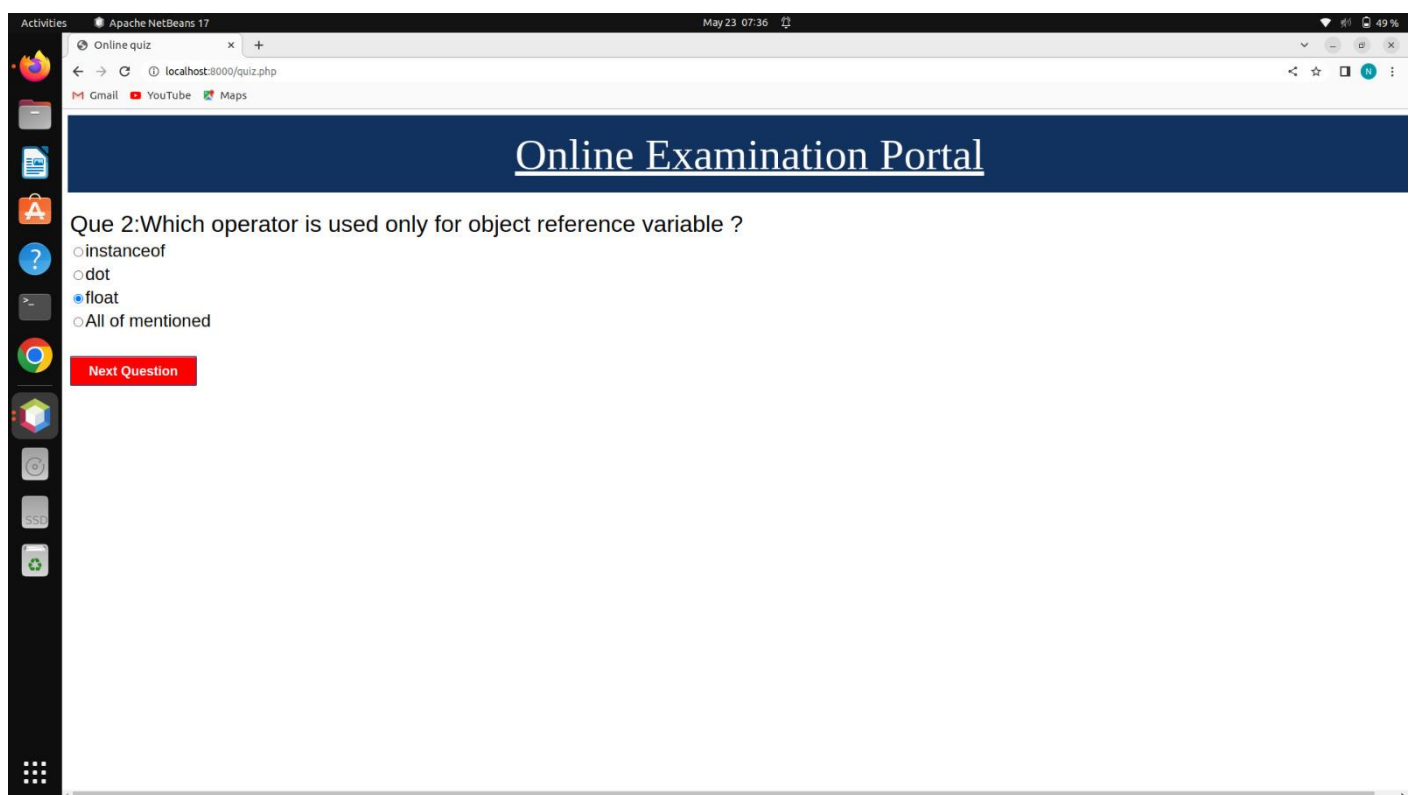
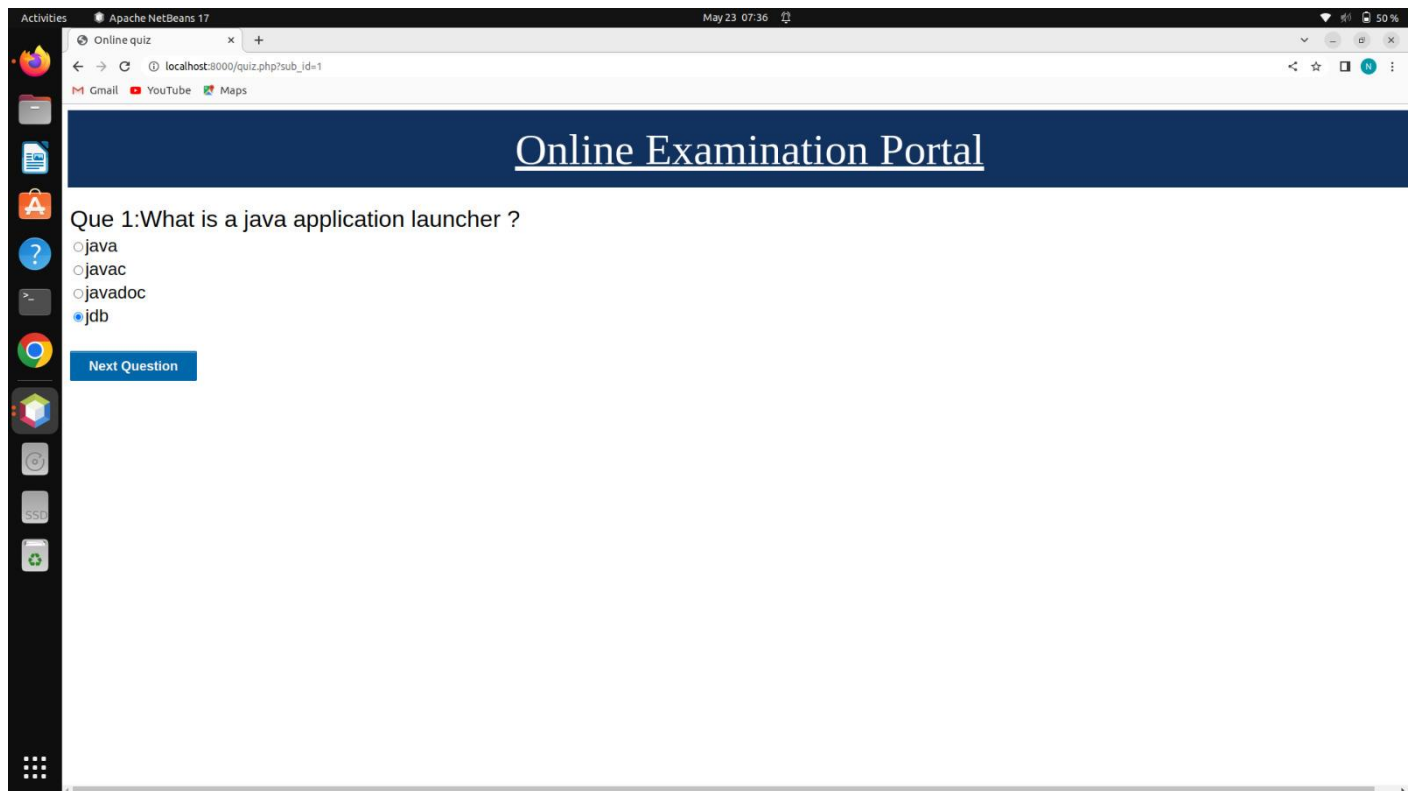
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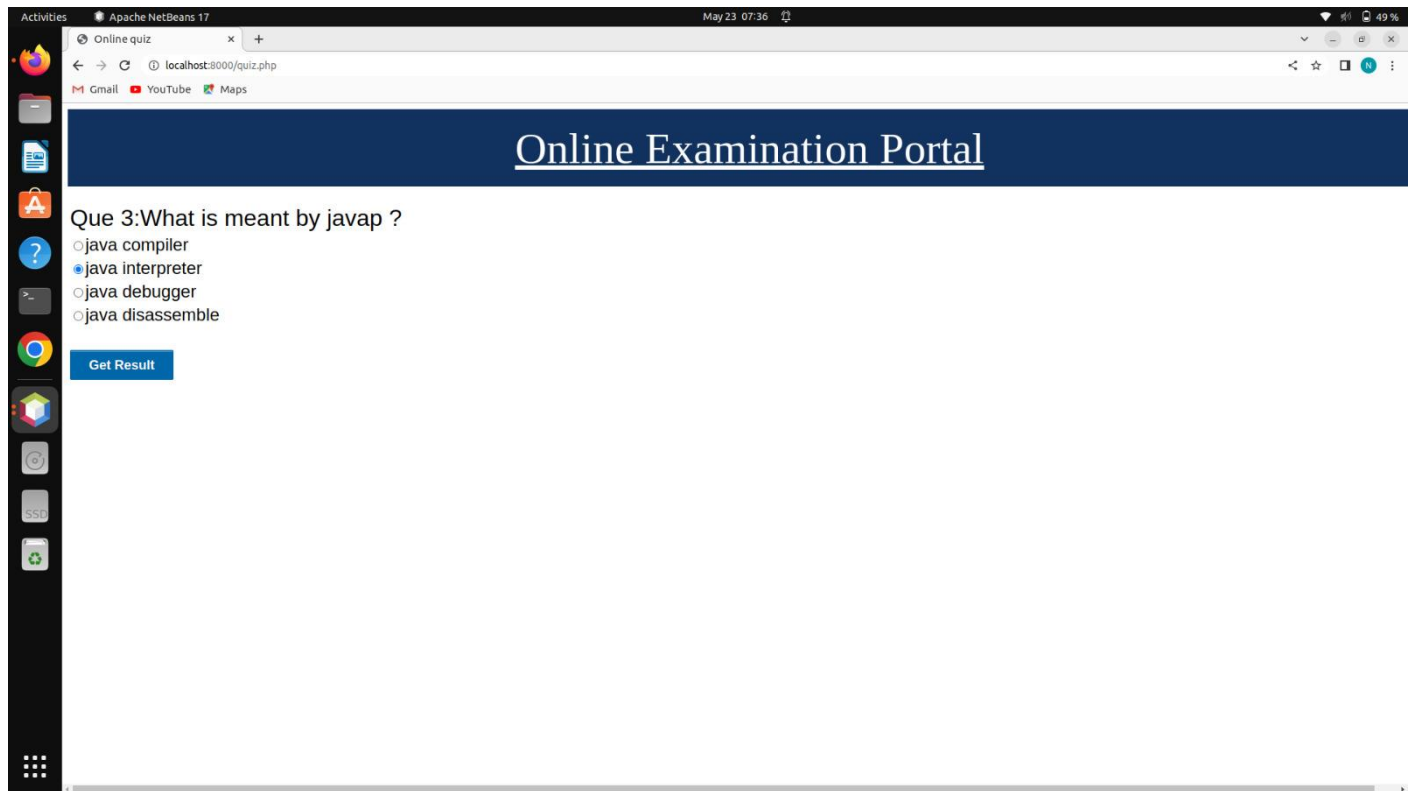
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Address :

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Report

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The problem with the current system is that student take exams manually. This outdated system will take long time utilization, the manual procedure used for conducting exam is much time-consuming process. Since covid-19 situation the existing system have many drawbacks like its time consuming. Results are not accurate since calculation is done manually.

Online Examination system saves the exams information in a database, and this make it an easier way to give exams. Students can give exam in a totally automated system.

6. System Testing

a. Defining Test Plan :

System Analysis is the process that decomposes a system into its component pieces for the purpose of defining how well those components interact to accomplish the set requirements. System design is the process of designing the elements of a system such as the architecture, modules and components, the different interfaces of those components and the data that goes through that system.

The purpose of the system design process is to provide sufficient detailed data and information about the system and its system elements to enable the implementation consistent with architectural entities as defined in models and views of the system architecture.

Elements of a System Architecture :

A test plan refers to a detailed document that catalogues the test strategy, objectives, schedule, estimations, deadlines, and the resources required for completing that project. Think of it as a blueprint for running the tests needed to ensure the software is working properly – controlled by test managers.

Modules – This are components that handle one specific tasks in a system. A combination of the modules makes up the system.

Components – This provides a particular function or group of related functions. They are Made up of modules.

Interfaces – This is the shared boundary across which the components of the system exchange information and relate.

Data – This is the management of the information and data flow.

b. Implementing Test Strategy :

The implementation of test strategy for an online examination system involves executing the planned testing activities to validate the system's functionality and performance and quality.

1. Unit Testing :

Unit testing of an online examination system involves testing individual units or components of the system in isolation to ensure their correctness and functionality. Unit Test includes identity units, defining test cases.

Identity Units :

Identify the individual units that needed to be tested. This include elements such as user registration, question submission, result calculation and reporting.

Defining Test Cases :

For User Login

Sr No.	Description	Input Data	Expected Result	Actual Result	Status
1.	Enter user ID in user ID Field	User ID	It should be accepted	Accepted	PASS
2.	Enter user ID in user ID Field	Wrong user ID	Error message should be given	Error message given	PASS
3.	Enter password in password Field	Password	It should accept	Accepted	PASS
4.	Enter password in password Field	Wrong password	Error message should be given	Error message given	PASS
5.	Login Button	Click	It should be accepted	Accepted	PASS

For User Registration

Sr No.	Description	Input Data	Expected Result	Actual Result	Status
1.	Enter first name in first name Field	First Name	It should be accepted	Accepted	PASS
2.	Enter first name in first name Field	Wrong first name	Error message should be given	Error message given	PASS
3.	Enter middle name in middle name Field	Middle name	It should accept	Accepted	PASS
4.	Enter middle name in middle name Field	Wrong middle name	Error message should be given	Error message given	PASS
5.	Enter last name in last name Field	last name	It should be accepted	Accepted	PASS
6.	Enter last name in last name Field	Wrong last name	Error message should be given	Error message given	PASS
7.	Enter user ID in user ID Field	User ID	It should be accepted	Accepted	PASS
8.	Enter user ID in user ID Field	Wrong user ID	Error message should be given	Error message given	PASS
9.	Enter password in password Field	Password	It should be accepted	Accepted	PASS
10.	Enter password in password Field	Wrong password	Error message should be given	Error message given	PASS
11.	Enter email in email Field	Email-id	It should be accepted	Accepted	PASS
12.	Enter wrong email in email Field	Wrong Email-id	Error message should be given	Error message given	PASS
13.	Register Button	Click	It should be accepted	Accepted	PASS

2. Black Box Testing :

Black Box Testing of an online examination system involves testing the system's functionality and behaviour without considering its internal implementation details. It focuses on validating the system's input and outputs based on its specified requirements. Black box testing includes requirements, design test cases, define test data, etc.

Requirements :

A clear understanding of the requirements and specifications of the online examination system. This includes user requirements, functional requirements, performance requirements and any constraint or limitations.

Design Test Cases :

Test cases should specify the input or actions to be performed, the expected output or system responses and any necessary constraints.

Define Test Data :

Identify and prepare necessary test data required for executing test cases. This includes user profiles, exam questions, predefined answers.

3. White Box Testing :

White Box Testing technique analyse the internal structure the used data structure, code structure etc. The working process of white box testing is requirements, functional requirement, design document and source code.

4. Alpha Testing :

Alpha Testing is a type of testing performed to identify bugs. Alpha Testing is one of the user acceptance testing.

7. Conclusion

- Our project is only a humble venture to satisfy the needs to manage their project work. Several user-friendly coding has also adopted. This package shall prove to be a powerful package in satisfying all the requirements of the Examination System.
- The objective of software planning is to provide a frame work that enables our team to make reasonable estimates made within a limited time frame at the beginning of the software project and should be updated regularly as the project progresses.
- So, for getting desire result within the time limits. The project can be completed with proper planning.

Future Plan

- We will host the platform on online servers to make it accessible worldwide.
- We can improve the services & we can add more Services.
- The project can be further extended by allowing the large database.

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